2.4.29

EE25BTECH11042 - Nipun Dasari

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Question

The points $\vec{A}(2, 9)$, $\vec{B}(a, 5)$ and $\vec{C}(5, 5)$ are the vertices of a triangle $\triangle A\vec{B}C$ right angled at $\vec{B}$.
Find the values of a and hence the area of $\triangle A\vec{B}C$.

Solution: Setting up the Vectors

Given the vertices:

$$\vec{A} = \begin{pmatrix} 2 \\ 9 \end{pmatrix}, \vec{B} = \begin{pmatrix} a \\ 5 \end{pmatrix}, \vec{C} = \begin{pmatrix} 5 \\ 5 \end{pmatrix} \tag{0.1}$$

We define the side vectors originating from the right-angled vertex $\vec{B}$.
Vector $\vec{c}$ (from $\vec{B}$ to $\vec{A}$):

$$\vec{c} = \vec{A} - \vec{B} = \begin{pmatrix} 2-a \\ 9-5 \end{pmatrix} = \begin{pmatrix} 2-a \\ 4 \end{pmatrix} \tag{0.2}$$

Vector $\vec{a}$ (from $\vec{B}$ to $\vec{C}$):

$$\vec{a} = \vec{C} - \vec{B} = \begin{pmatrix} 5-a \\ 5-5 \end{pmatrix} = \begin{pmatrix} 5-a \\ 0 \end{pmatrix} \tag{0.3}$$

Solution: Finding 'a' using Orthogonality

Since the triangle is right-angled at $\vec{B}$, the vectors $\vec{c}$ and $\vec{a}$ are orthogonal. Their inner product must be zero.

$$\vec{c}^T \vec{a} = \begin{pmatrix} 2-a & 4 \end{pmatrix} \begin{pmatrix} 5-a \\ 0 \end{pmatrix} = 0 \tag{0.4}$$

Solving the equation:

$$(2-a)(5-a) + (4)(0) = 0 \tag{0.5}$$
$$(2-a)(5-a) = 0 \tag{0.6}$$

This gives two possible values: $a = 2$ or $a = 5$.
If $a = 5$, vertex $\vec{B}$ becomes $(5, 5)$, which is the same as vertex $\vec{C}$. This would not form a triangle.
Therefore, we must have $a = 2$.

Solution: Calculating the Area (Setup)

With $a = 2$, the side vectors are:

$$\vec{c} = \begin{pmatrix} 2-2 \\ 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 4 \end{pmatrix} \tag{0.7}$$

$$\vec{a} = \begin{pmatrix} 5-2 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \end{pmatrix} \tag{0.8}$$

The area is calculated using the cross product formula:

$$\Delta = \frac{1}{2}\|\vec{c} \times \vec{a}\| \tag{0.9}$$

In 3D space, the vectors are:

$$\vec{c} = \begin{pmatrix} 0 \\ 4 \\ 0 \end{pmatrix} \qquad \vec{a} = \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} \tag{0.10}$$

Solution: Calculating the Cross Product

Using the component-wise formula $\vec{A} \times \vec{B} = \begin{pmatrix} |\vec{A}_{23} & \vec{B}_{23}| \\ |\vec{A}_{31} & \vec{B}_{31}| \\ |\vec{A}_{12} & \vec{B}_{12}| \end{pmatrix}$:

$$|\vec{c}_{23}\ \vec{a}_{23}| = \begin{vmatrix} 4 & 0 \\ 0 & 0 \end{vmatrix} = 0 \tag{0.11}$$

$$|\vec{c}_{31}\ \vec{a}_{31}| = \begin{vmatrix} 0 & 0 \\ 0 & 3 \end{vmatrix} = 0 \tag{0.12}$$

$$|\vec{c}_{12}\ \vec{a}_{12}| = \begin{vmatrix} 0 & 4 \\ 3 & 0 \end{vmatrix} = 0 - 12 = -12 \tag{0.13}$$

The cross product vector is:

$$\vec{c} \times \vec{a} = \begin{pmatrix} 0 \\ 0 \\ -12 \end{pmatrix} \tag{0.14}$$

The norm is $\|\vec{c} \times \vec{a}\| = \sqrt{0^2 + 0^2 + (-12)^2} = 12$.

Solution: Final Area

Substituting the norm into the area formula:

$$\Delta = \frac{1}{2}\|\vec{c} \times \vec{a}\| = \frac{1}{2}(12) = 6 \tag{0.15}$$

The area of the triangle is 6 square units.

Plot 1

Figure_1.png

Plot 2

Figure_2.png